LOGIC BUILDING WITH LOOPS

(BEFORE STARTING DSA)

Phase 1 – While Loop
Phase 2 – do while loop
Phase 3 – For Loop
Phase 4 – Nested loop Logic
Phase 5 – Break / Continue logic
Phase 6 – Mathematical Series
Phase 7 – Mixed Logical Loop Problems
Phase 8 – Star Pattern Printing

Repositories for Answers:

- **C:** logic-building-codeN-C
- **C++:** logic-building-codeN-Cpp
- **Java:** logic-building-codeN-Java
- **Python:** logic-building-codeN-Python
- **JavaScript:** logic-building-codeN-JavaScript

Phase 1 : While Loop

1. Print all numbers from **1 to 10** using a loop.
2. Print numbers from **10 down to 1** in reverse order.
3. Print **all even numbers** between **1 and 100**.
4. Print **all odd numbers** between **1 and 100**.
5. Print the **multiplication table** of a given number from **n × 1 to n × 10**.
6. Calculate and print the **sum of the first n natural numbers**.
7. Calculate the **sum of all even numbers** from **1 up to n**.
8. Calculate the **sum of all odd numbers** from **1 up to n**.
9. Calculate and print the **factorial of a given number**.
10. Find and print the **product of all digits** of a given number.
11. Count and print the **total number of digits** in a given number.
12. Reverse the given number and print the **reversed value**.
13. Check whether the given number is a **palindrome**.
14. Find and print the **sum of digits** of the given number.
15. Check whether the given number is an **Armstrong number**.
16. Check whether the given number is a **Perfect number**.
17. Print **all prime numbers** between **1 and 100**.
18. Check whether the given number is a **prime number**.
19. Print the **Fibonacci series** up to **n terms**.
20. Find and print the **sum of the Fibonacci series** up to **n terms**.
21. Print the **square of each number** from **1 to n**.
22. Print the **cube of each number** from **1 to n**.

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23. Print all numbers between **a and b** that are **divisible by 7**.
24. Print **all factors** of the given number.
25. Find and print the **sum of all factors** of the given number.
26. Find the **HCF (Highest Common Factor)** of two given numbers.
27. Find the **LCM (Least Common Multiple)** of two given numbers.
28. Find the **smallest digit** in the given number.
29. Find the **largest digit** in the given number.

Phase 2 : do - While Loop

1. Print all numbers from **1 to 10**.
2. Print the **multiplication table** of a given number.
3. Keep taking numbers from the user until **0 is entered**, then print the **sum of all entered numbers**.
4. Keep taking numbers from the user until **0 is entered**, then print the **largest number** among all inputs.
5. Count and print the **number of digits** in the given number.
6. Reverse the given number and print the **reversed value**.
7. Check whether the given number is a **palindrome**.
8. Check whether the given number is an **Armstrong number**.
9. Calculate and print the **factorial** of the given number.
10. Print the **Fibonacci series** up to the required number of terms.
11. Find the **HCF (Highest Common Factor)** of the given numbers.
12. Create a **menu-driven program** that allows the user to choose and perform different operations.
13. Keep taking numbers from the user until a **negative number is entered**, then count how many **positive numbers** were entered.
14. Find and print the **sum of digits** of the given number.
15. Calculate and print the **sum of even digits** and the **sum of odd digits** of the given number separately.

Phase 3 : For Loop

1. Print all numbers from **1 to 10**.
2. Print numbers from **10 down to 1** in reverse order.
3. Print **all even numbers** between **1 and 100**.
4. Print **all odd numbers** between **1 and 100**.
5. Print the **multiplication table** of a given number.
6. Calculate and print the **factorial of a given number**.
7. Calculate and print the **factorial of every number** from **1 to n**.
8. Print **all prime numbers** between **1 and 100**.
9. Check whether the given number is a **prime number**.

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10. Print the **Fibonacci series** up to the required number of terms.
11. Find and print the **sum of the Fibonacci series**.
12. Print **all factors** of the given number.
13. Find and print the **sum of all factors** of the given number.
14. Find the **HCF (Highest Common Factor)** of the given numbers.
15. Find the **LCM (Least Common Multiple)** of the given numbers.
16. Print the **square of each number** from **1 to n**.
17. Print the **cube of each number** from **1 to n**.
18. Print all numbers between **a and b** that are **divisible by 7**.
19. Find and print the **sum of the first n natural numbers**.
20. Find and print the **sum of all even numbers** from **1 up to n**.
21. Find and print the **sum of all odd numbers** from **1 up to n**.

Phase 4 : Nested Loop Logic

1. Print the **multiplication tables for all numbers from 1 to 10**.
2. Print **all possible pairs (i, j)** where both `i` and `j` range from **1 to n**.
3. For every number from **1 to n**, count and print the **total number of its factors**.
4. Print **all prime numbers** up to n using **nested loop checking**.
5. Print the **Fibonacci pattern row by row**, where each row prints the next Fibonacci numbers
6. Generate and print a **number triangle pattern** using **nested loops**.
7. Print a **matrix**, then calculate and display the **sum of each row** and the **sum of each column**.
8. Print **all Pythagorean triplets** whose values are **less than or equal to n**.

Phase 5 : Break / Continue Logic

1. Print numbers from **1 to 100**, and **stop the loop** as soon as a number **divisible by 17** is encountered.
2. Print numbers from **1 to 100**, but **skip** all numbers that are **divisible by 5** and continue printing the rest.
3. Take **5 numbers** as input, **skip any number that is 0** using `continue`, and calculate the **sum of the remaining numbers**.
4. Search for a specific number in a list of inputs, and **terminate the loop** immediately when the number is **found**.
5. Keep taking numbers from the user and **print them until a negative number appears**, then stop the loop.
6. Skip all **odd numbers** and print **only the even numbers**.
7. Continuously add numbers in a loop and **stop the loop** when the **sum becomes greater than 100**

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19.

```
    A
   BCD
  EFGHI
 JKLMNOP
QRSTUVWXY
```

20.

```
1
12
123
1234
12345
```

21.

```
    1
   121
  12321
 1234321
123454321
```

22.

```
*
**
****
*****
****
***
**
*
```

23.

```
*
**
***
****
*****
*****
****
***
**
*
```

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